

SafeSpace: Stress Detection and Management System for Young Professionals

1 Introduction

In a world where demands on young professionals and students are escalating, mental health has become a pressing issue. Academic and job pressures, coupled with personal responsibilities, contribute to rising levels of stress and anxiety. The World Health Organization estimates that 7.5% of Indians currently face mental health challenges, with projections suggesting this could increase to 20% by year-end. Deloitte’s survey reveals that 75% of young professionals experience high stress, often leading to job changes due to mental health concerns. This scenario underscores the urgent need for innovative mental health solutions.[1]. These statistics underscore the pressing need for effective mental health management strategies. The rising prevalence of stress and its detrimental effects on well-being necessitate innovative solutions to help individuals manage their mental health effectively. Recognizing this urgent need, our research introduces SafeSpace, a cutting-edge stress management platform tailored specifically for young professionals. SafeSpace integrates advanced technologies such as image analysis, Natural language processing (NLP), and integrated hardware sensors to provide real-time stress detection and personalized management strategies.

SafeSpace employs convolutional neural networks (CNNs)[2] for sophisticated image analysis, detecting subtle signs of stress such as changes in facial expressions and posture. Additionally, NLP models analyze text and voice inputs to gauge users’ emotional states, offering tailored recommendations and interventions. Hardware sensors track physiological indicators like heart rate and skin conductivity, enhancing the platform’s accuracy. As stress and mental health issues rise among young professionals and students, SafeSpace offers a significant advancement in stress management technology, combining advanced image analysis, NLP, and sensor data for real-time, personalized mental health care.

SafeSpace’s user-friendly Android app integrates these technologies seamlessly, enabling easy access to real-time stress management tools. By providing personalized feedback and proactive interventions, SafeSpace helps users manage stress effectively, fostering a healthier and more balanced lifestyle.

1.1 Motivation

In today’s fast-paced world, mental health has emerged as a critical concern, particularly among young professionals and students. The pressures of academic performance, professional achievements, and personal responsibilities contribute significantly to stress, anxiety, and other mental health issues. According to the World Health Organization (WHO), approximately 7.5% of Indians suffer from mental disorders, a figure expected to rise to 20% by the end of the year. In the workplace, a survey by Deloitte revealed that nearly 75% of young professionals report experiencing substantial stress, with many leaving jobs due to mental health reasons (Deloitte, 2020) [1]. These statistics underscore the pressing need for effective mental health management strategies. The rising prevalence of stress and its detrimental effects on well-being necessitate innovative solutions to help individuals manage their mental health effectively.

1.2 Contributions

Recognizing this urgent need, our research introduces SafeSpace, a cutting-edge stress management platform tailored specifically for young professionals. The contributions of this research are as follows:

- i. **Integration of Advanced Technologies:** SafeSpace combines image analysis, Natural Language Processing (NLP), and integrated hardware sensors to provide real-time stress detection and personalized management strategies.
- ii. **Sophisticated Image Analysis:** The platform employs CNNs [3, 4, 5] for sophisticated image analysis, enabling the detection of subtle physical signs of stress, such as changes in facial expressions, posture, and other physiological indicators. This allows for accurate and real-time feedback.
- iii. **Advanced NLP Models:** SafeSpace utilizes advanced NLP models to analyze text [6] and voice inputs from users, interpreting the emotional tone and content of communications to gain further insights into users' mental states. This enables the platform to offer personalized recommendations and interventions based on individual needs and emotional context.
- iv. **Integration of Hardware Sensors:** The platform includes hardware sensors to monitor physiological parameters such as heart rate, skin conductivity, and body temperature, which are critical indicators of stress levels. Combining data from these sensors with image and text analysis provides a comprehensive and accurate assessment of users' stress levels.[7]
- v. **User-Friendly Application Interface:** SafeSpace is integrated into an intuitive Android application interface, designed to ensure ease of access and utilization of its features. This enhances user experience and engagement.
- vi. **Personalized Recommendations and Proactive Interventions:** By delivering tailored recommendations and proactive interventions, SafeSpace empowers users to manage their mental health proactively, improving overall well-being and productivity. The platform's real-time capabilities help users address stress before it becomes overwhelming.
- vii. **Continuous Monitoring and Feedback:** SafeSpace supports users in developing effective coping strategies and building resilience against stress through continuous monitoring and personalized feedback.

2 Literature Review

Kanaparthi *et al.* [7] presented a machine learning-based approach to identify stress levels in IT employees, aiming to enhance workplace well-being and productivity through accurate stress detection methods. Almeida *et al.* [3] proposed a deep learning-based facial expression recognition system for stress detection. The VGG16 model presented an accuracy of 92.1%. Mohammadpour *et al.* [4] aims to enhance accuracy in identifying emotions from facial expressions using advanced CNN techniques. The experimental results show that deep CNNs lead to an increase in facial emotion recognition accuracy *i.e.* 97.01% as compared to traditional CNN *i.e.* 95.75%. Cohen *et al.* [8] proposed a method for emotion recognition from facial expressions using Multilevel Hidden Markov Models (HMM), aiming to capture nuanced emotional states with improved accuracy. The results revealed the superiority of multilevel HMM as compared to one layered HMM. Li *et al.* [9] comprehensively reviewed advancements in deep learning techniques for facial expression recognition, emphasizing recent developments and challenges in the field. A comparative analysis on widely used benchmarks, challenges, opportunities and future directions for designing robust facial expression recognition systems is also presented. Solis-Arrazola *et al.* [10] proposed a novel approach to improve image-based facial expression recognition by extracting facial features based on muscle activation patterns, aiming to enhance accuracy and robustness in emotion detection. Solis-Arrazola *et al.* [10] proposed a novel approach to improve image-based facial expression recognition by extracting facial features based on muscle activation patterns, aiming to enhance accuracy and robustness in emotion detection. The study suggested that FMA-based approaches could be effective for facial expression classification across a range of age groups and genders. Rastogi *et al.* [6] introduced a new benchmark dataset and conducts an analytical study on stress detection from social media articles, presented at the 2022 International Joint Conference on Neural Networks. Chew *et al.* [5] proposed an Enhanced Stress CNN for facial expression recognition aimed at stress detection, offering advancements in accuracy and performance in this application domain. ESCNN achieves the best result, which is 95% for Stress Recognition Accuracy (15 epoch) and 85.54% accuracy of Facial Expression (15 epoch), as compared with ResNet 50 (81.67%) and Sequential Model (77.32%). Overall, ESCNN is viable for stress detection and recognition of facial expressions. Rachakonda *et al.* [11] integrated IoMT and blockchain to develop an automated stress detection system, monitoring physiological parameters like pulse rate and blood oxygen levels to assess and manage stress non-invasively. The proposed SaYoPillow not only collects and analyzes eight different physiological signal data to predict stress, but also educates the user

about the benefits of smart-sleeping. A brief summary of existing works for stress detection is given in Table 1.

Table 1: Overview of existing works for stress detection

Author	Model Used	Inference	Dataset Used
Kanaparthi <i>et al.</i> [7]	KNNs	Implemented stress detection using KNNs and transfer learning for better accuracy	DEAP [12] and RAVDESS [13] datasets
Almeida <i>et al.</i> [3]	VGG16, VGG19, and Inception-ResNet V2	Leveraged CNNs for stress detection, improving classification of facial expressions	KDEF Dataset [14]
Mohammadpour <i>et al.</i> [4]	CNNs	Used deep CNNs for accurate emotion recognition, enhancing our recognition system	CK+[15] and JAFFE[16] Datasets
Cohen <i>et al.</i> [8]	HMMs	Applied multilevel HMMs for emotion recognition, aiding in modeling emotional states	JAFFE[16], YALE multimodal [17] Datasets and more
Li <i>et al.</i> [9]	Survey	Surveyed deep learning techniques for facial expression recognition, guiding our approach	FER 2013[18], CK+[15] and more
Solis-Arrazola <i>et al.</i> [10]	Custom	Enhanced recognition via muscle activation-based features, improving feature extraction	FACES[19], DEF[20] and JAFFE[16] datasets
Rastogi <i>et al.</i> [6]	Benchmark	Introduced stress detection from social media, providing a new dataset benchmark	Self-constructed datasets: Twitter Full and Non-Advert, Reddit Title and Combi
Chew <i>et al.</i> [5]	Enhanced CNNs	Developed enhanced CNNs for stress detection, boosting recognition accuracy	FER2013[18]
Rachakonda <i>et al.</i> [11]	IoMT	Integrated blockchain for IoMT framework, ensuring privacy in stress management	SaYoPillow Dataset [21]
Mohammad <i>et al.</i> [22]	Mechanical Turk Human Intelligence Tasks (HITS)	Shows efficient crowdsourcing to create large, high-quality word-emotion and word-polarity lexicons	EmoLex [23]

3 Data Collection

For training, images from the Karolinska Directed Emotional Faces (KDEF)[14] dataset were utilized. The KDEF dataset is recognized for its diverse collection of facial expressions, providing a suitable foundation for training machine learning models in emotion recognition tasks. For testing, the Cohn-Kanade (CK+)[15] and Taiwanese Face Expression Image Database (TFEID) [24] datasets were employed. These datasets are well-established benchmarks in the field, known for their large-scale, annotated collections of facial expressions captured under controlled conditions. The Taiwanese Face Expression Image Database (TFEID) images, initially in grayscale, were converted to color format for experimentation and testing purposes. Figure 1 shows the CK+ dataset, initially in grayscale, converted to color for testing.

Karolinska Directed Emotional Faces (KDEF) [14] dataset includes 4900 standardized images of human facial expressions across seven emotions from 70 models, crucial for facial expression recognition and emotion analysis in psychology and computer vision. Cohn-Kanade (CK+) [15] features 593 image sequences capturing basic emotions from 123 subjects with detailed annotations, aiding precise research in facial expression recognition. Taiwanese Face Expression Image Database (TFEID) [25] comprises 2400 images from 24 subjects showcasing various expressions in grayscale and color, managed by Academia Sinica for studying human facial expressions. The TFEID images, initially in grayscale, were converted to color format for experimentation and testing purposes. Figure 1 shows the CK+ dataset, initially in grayscale, converted to color for testing. SaYoPillow [21] dataset by Rachakonda Kondal, developed in 2024, focuses on detecting human stress through sleep patterns.

The custom dataset for Stress Management comprises emotion vectors linked to specific YouTube videos. The dataset is curated to ensure diverse emotional content and relevant video suggestions.

The SVM classifier and KNN model are trained using this dataset to achieve accurate and meaningful recommendations.

4 Proposed Methodology

4.1 Stress Detection

Stress detection involves identifying physiological and emotional indicators to assess an individual's stress levels. In the following sections, we will discuss the methodologies used for real-time stress detection, evaluate the accuracy and effectiveness of these methods, and explore their practical applications and potential improvements.

4.1.1 CNN-ELM Architecture

The proposed CNN-ELM approach combines CNN and ELM for facial emotion recognition task. The configuration used for both the models is as described below:

- **ELM:** ELM model is configured with 100 hidden neurons and employs hyperbolic tangent ($\tanh$) activation function. ELMs were selected for their computational efficiency and effectiveness in handling high-dimensional feature spaces.
- **CNN:** CNN architecture is based on ResNet-50, and pre-trained on the ImageNet dataset. The base model includes layers such as GlobalAveragePooling2D for spatial aggregation, ReLU activation functions for feature extraction, a Dropout layer (dropout rate of 0.5) to mitigate overfitting, and a final softmax activation layer for multi-class classification.
- **CNN Compilation:** CNN model is compiled using Stochastic Gradient Descent (SGD) optimizer with a specific learning rate tailored to the task. The categorical cross-entropy loss function is chosen for its suitability in multi-class classification scenarios such as facial emotion recognition.

By integrating CNN with ELM, the study aims to enhance the accuracy and efficiency of facial emotion recognition systems. The adoption of deep learning techniques alongside efficient learning mechanisms such as ELMs holds promise for advancing automated emotion detection capabilities from facial images.

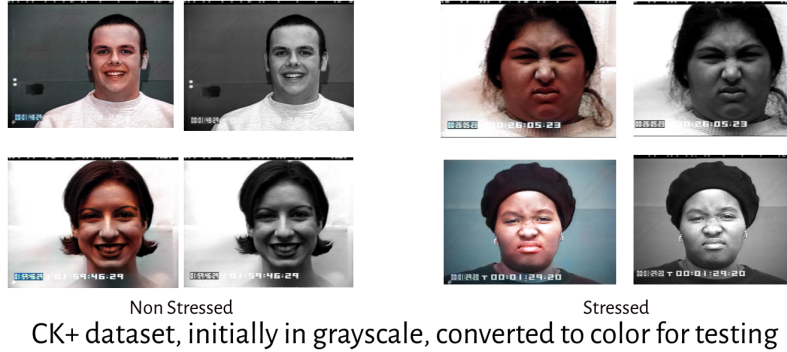


Fig. 1: Dataset images converted to grayscale showing stressed and non-stressed facial expressions.

4.2 Stress Classification

The classification methodology involves utilizing sensor data to determine stress levels on a scale from 0 to 4. This section details the systematic approach taken to integrate and utilize the selected sensors and cloud infrastructure.

4.2.1 Sensor Selection and Initial Implementation

The project focused on selecting appropriate sensors based on criteria such as functionality, cost-effectiveness, and ease of integration. The Adafruit MLX90614 and MAX30100 sensors were chosen

for their capabilities in non-contact temperature measurement and pulse rate/SpO2 monitoring, respectively. These sensors operate at I2C addresses 0x5A and 0x57, respectively.

4.2.2 Integration of Sensors

Initial implementation included constructing individual circuits for each sensor to ensure they operated according to specifications. Challenges arose during integration, particularly in synchronizing data acquisition due to differences in baud rates. Solutions involved meticulous adjustment of communication protocols to achieve synchronized operation within a unified circuit.

4.2.3 Interfacing with ESP WROOM 32 WiFi Module

To enable wireless transmission of sensor data for remote monitoring and analysis, the integrated sensor circuitry interfaced with an ESP WROOM 32 WiFi module. Direct connections encountered technical obstacles related to simultaneous Serial Clock Line (SCL) requirements. These challenges were resolved by utilizing an Arduino Uno board to manage signal conflicts and ensure seamless integration with the ESP module.

4.2.4 Cloud Storage Platform Selection

Selection of a suitable cloud storage platform was critical for real-time data logging and analysis. Initially utilizing the Blynk IoT platform for real-time data visualization, the project transitioned to Firebase Realtime Database for its comprehensive capabilities in storing and retrieving historical sensor data. As shown in Figure 2, the Firebase Realtime Database interface...

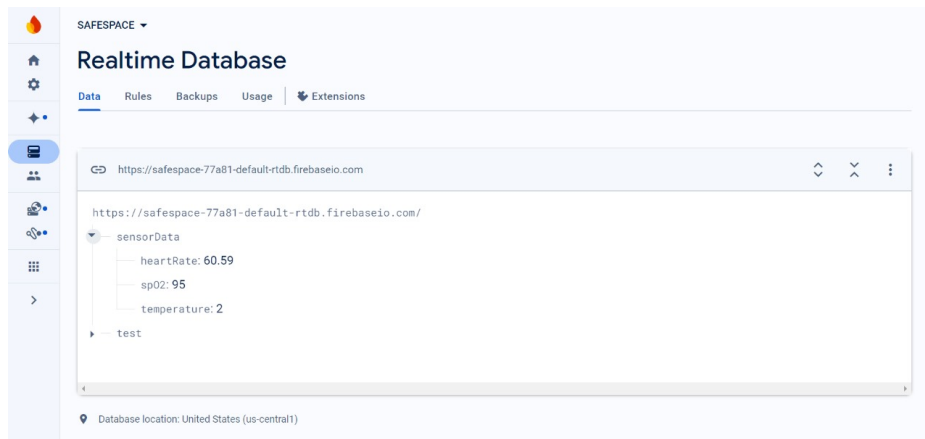


Fig. 2: Firebase Realtime Database Interface

4.2.5 Implementation Summary

- **Sensor Selection:** Adafruit MLX90614 for non-contact temperature measurement and MAX30100 for pulse rate/SpO2 monitoring.
- **Circuit Integration:** Individual circuits validated and integrated into a unified system, overcoming baud rate discrepancies.
- **ESP Module Integration:** Arduino Uno employed to manage SCL conflicts, enabling successful interfacing with the ESP WROOM 32 WiFi module.
- **Cloud Storage:** Transition from Blynk IoT to Firebase Realtime Database for robust real-time and historical data storage capabilities.

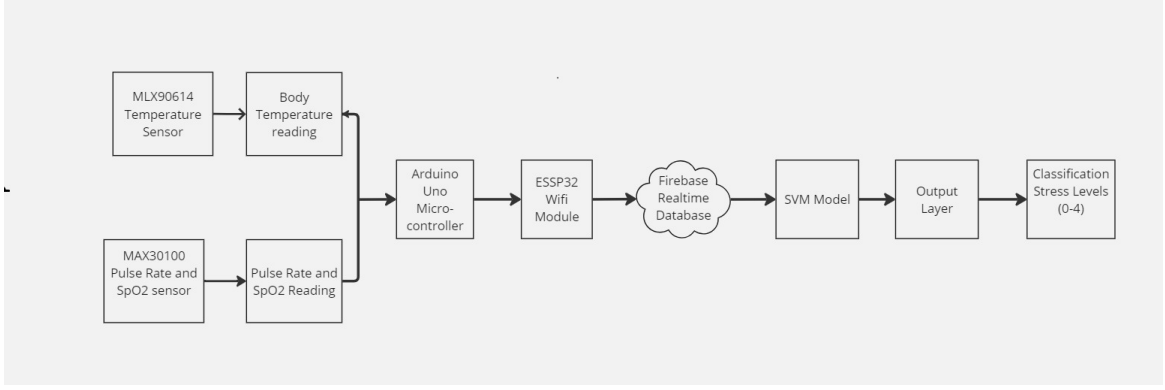


Fig. 3: Circuit Block diagram of hardware component for stress detection

This methodology outlines the systematic approach taken to select, implement, and integrate hardware components and interface them with cloud infrastructure to classify stress levels from 0 to 4 based on sensor data.

4.3 Stress Management

This section outlines the development and implementation of a recommendation system designed to analyze user text input and suggest relevant YouTube videos. The system utilizes natural language processing (NLP) techniques and machine learning models to provide personalized recommendations aimed at improving student well-being and managing stress.

4.3.1 Recommendation System Process

The recommendation system processes user text through several stages to generate video suggestions:

1. **Text Preprocessing:**

- **Part of Speech Tagging:** Identifies the grammatical components of the text.
- **Lemmatization:** Converts words to their base forms.
- **Stop Word Removal:** Eliminates common words that do not contribute significant meaning.
- **Tokenization:** Splits the text into individual tokens (words or phrases).

2. **Emotion Vector Conversion:** The preprocessed text is transformed into a vector using the NRC Emotion Lexicon. This lexicon includes annotations for eight basic emotions (anger, fear, anticipation, trust, surprise, sadness, joy, and disgust) and sentiments (positive and negative). The lexicon assigns emotional weights to words, enabling the system to quantify the emotional and sentimental content of the text.

3. **SVM Classification:** The emotion vector is inputted into a Support Vector Machine (SVM) classifier. The SVM is trained on a custom dataset that pairs emotion vectors with links to YouTube videos. The classifier predicts a single video link that best matches the emotional profile of the user's text.

4. **KNN-Based Video Expansion:** The output video link from the SVM classifier is then used as input for a K-Nearest Neighbors (KNN) unsupervised model. This model identifies 6 additional similar videos, expanding the range of recommendations and providing users with a variety of options.

5 Experimental Results and Analysis

5.1 Performance Evaluation

5.1.1 Stress Detection

To assess the performance of our image stress detection model, we evaluated it using various parameters, including loss, accuracy, and confusion matrix during both the training and validation phases. The following sections provide a detailed analysis of these parameters.

Epoch vs. Loss

The training loss over epochs is depicted in Fig. 4. The loss curve provides insight into the model's learning process. A steadily decreasing loss indicates that the model is learning effectively.

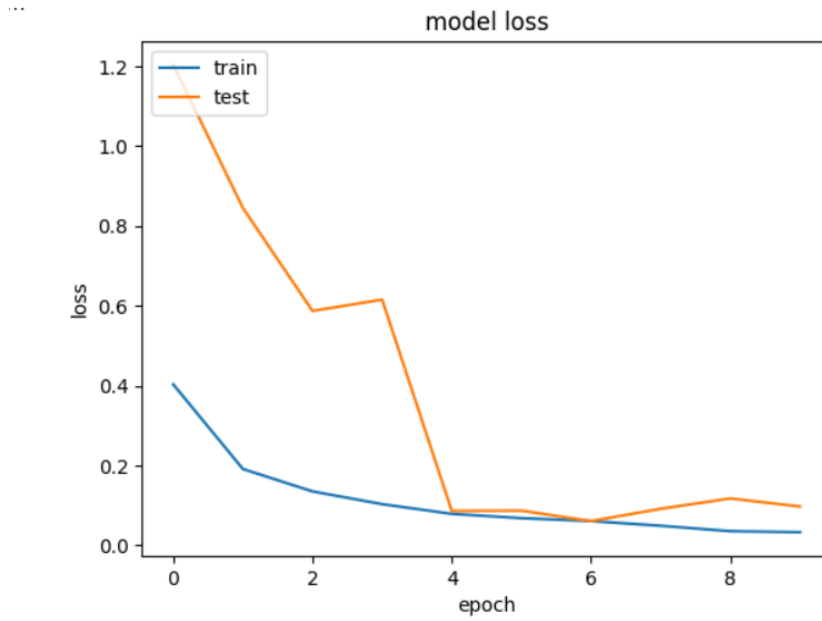


Fig. 4: Variation of training loss with number of epochs

Epoch vs. Accuracy

The accuracy of the model during the training phase over different epochs is illustrated in Fig. 5. A high and stable accuracy on the training set indicates that the model is effectively learning the training data.

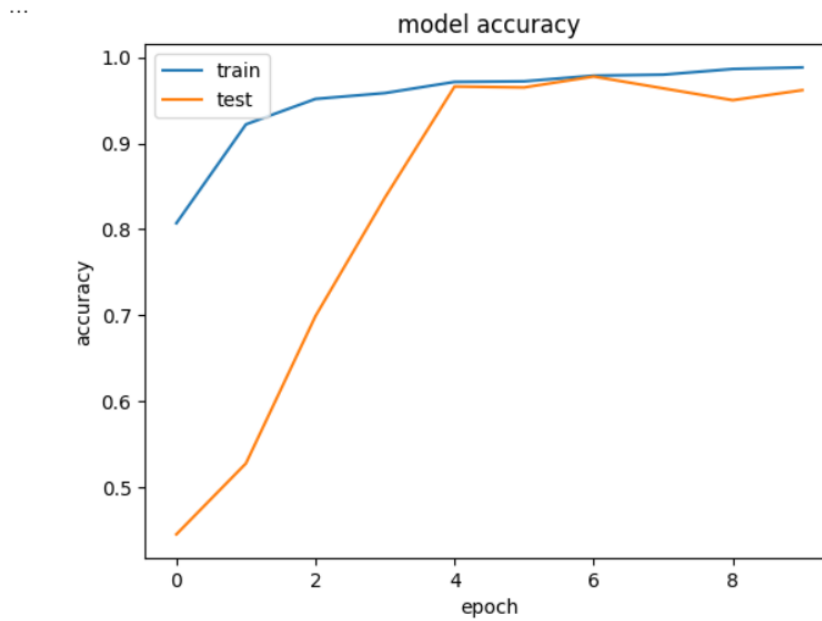


Fig. 5: Variation of training accuracy with number of epochs

Confusion Matrix

The confusion matrix in Fig. 6 offers a detailed breakdown of the model’s performance by displaying the true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN). This matrix helps in understanding the model’s capability to distinguish between “stressed” and “not stressed” accurately during the training phase.

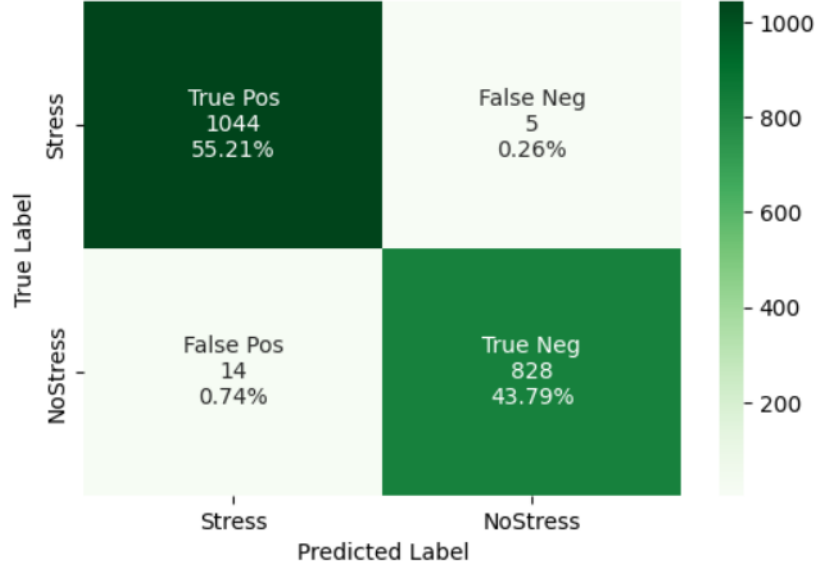


Fig. 6: Confusion matrix for the image stress detection model (training phase)

Performance Metrics

From the confusion matrix, we can derive various performance metrics such as accuracy, precision, recall, and F1-score. These metrics provide a quantitative measure of the model’s effectiveness in stress detection during training. The equations for these metrics are as follows:

- **Accuracy:** It is computed as the ratio of correctly classified instances to the total number of instances as given by Equation 1.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

- **Precision:** It signifies the proportion of true positive predictions out of the total positive predictions as given by Equation 2.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2)$$

- **Recall (Sensitivity):** It defines the ability of the model to identify all relevant instances of a particular class as given by Equation 3.

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3)$$

- **F1-Score:** It is computed as the harmonic mean of precision and recall, providing a balanced measure of both metrics as given by Equation 4.

$$\text{F1-Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

Validation Results

In this section, we present the performance metrics of different classifiers evaluated on two datasets: CK+ and KDEF. The following tables summarize the key results, including precision, recall, F1-score, accuracy, and other relevant metrics.

Classifier Performance on CK+ and KDEF. Table 2 displays the performance metrics for various classifiers trained and tested on the CK+ and KDEF datasets.

Table 2: Performance Metrics of Classifiers on CK+ and KDEF Datasets

Metric	Training on CK+	Training on KDEF
Random Forest	Training Time: 9.37s Testing Time: 0.07s Precision: 0.89 Recall: 0.69 F1-Score: 0.78 Accuracy: 0.85 AUC Score: 0.82 Specificity: 0.95	Training Time: 23.97s Testing Time: 0.06s Precision: 0.43 Recall: 0.18 F1-Score: 0.25 Accuracy: 0.59 AUC Score: 0.51 Specificity: 0.85
KNN	Training Time: 0.08s Testing Time: 2.20s Precision: 0.86 Recall: 0.64 F1-Score: 0.73 Accuracy: 0.82 AUC Score: 0.79 Specificity: 0.93	Training Time: 0.19s Testing Time: 4.31s Precision: 0.27 Recall: 0.36 F1-Score: 0.31 Accuracy: 0.37 AUC Score: 0.37 Specificity: 0.37
DT	Training Time: 0.07s Testing Time: 0.03s Precision: 0.62 Recall: 0.41 F1-Score: 0.49 Accuracy: 0.68 AUC Score: 0.63 Specificity: 0.84	Training Time: 0.19s Testing Time: 0.03s Precision: 0.36 Recall: 0.80 F1-Score: 0.50 Accuracy: 0.38 AUC Score: 0.46 Specificity: 0.11
LDA	Training Time: 25.40s Testing Time: 0.05s Precision: 0.90 Recall: 0.70 F1-Score: 0.79 Accuracy: 0.85 AUC Score: 0.83 Specificity: 0.95	Training Time: 99.89s Testing Time: 0.05s Precision: 0.50 Recall: 0.51 F1-Score: 0.50 Accuracy: 0.61 AUC Score: 0.59 Specificity: 0.68
SVM	Testing Time: 42.56s Precision: 0.98 Recall: 0.90 F1-Score: 0.94 Accuracy: 0.95 AUC Score: 0.94 Specificity: 0.99	Training Time: 455.85s Testing Time: 96.05s Precision: 0.38 Recall: 0.60 F1-Score: 0.47 Accuracy: 0.47 AUC Score: 0.49 Specificity: 0.39
GNB	Training Time: 0.68s Testing Time: 0.98s Precision: 0.64 Recall: 0.71 F1-Score: 0.67 Accuracy: 0.73 AUC Score: 0.73 Specificity: 0.75	Training Time: 1.69s Testing Time: 0.97s Precision: 0.54 Recall: 0.35 F1-Score: 0.42 Accuracy: 0.63 AUC Score: 0.58 Specificity: 0.81
ELM	Training Time: 0.04s Testing Time: 0.005s Precision: 0.85 Recall: 0.81 F1-Score: 0.84 Accuracy: 0.83	Training Time: 0.04s Testing Time: 0.005s Precision: 0.87 Recall: 0.83 F1-Score: 0.84 Accuracy: 0.86

Impact of Sigma on Performance. Table 3 shows the impact of varying the sigma parameter on the performance metrics for the classifiers. It illustrates the impact of different sigma values on various performance metrics. As the sigma parameter changes, different aspects such as training time, precision, and accuracy are affected, providing insights into how sigma influences model performance. The results are detailed for different sigma values, covering training time, testing time, precision, recall, F1-score, and accuracy.

Table 3: Impact of Sigma on Performance Metrics

Metric	Sigma 100	Sigma 150	Sigma 200	Sigma 250	Sigma 300	Sigma 350	Sigma 400
Training Time (CK+)	0.019	0.024	0.025	0.028	0.030	0.036	0.040
Testing Time (CK+)	0.005	0.006	0.010	0.010	0.010	0.011	0.012
Precision (CK+)	0.986	0.986	0.983	0.989	0.990	0.990	0.982
Recall (CK+)	0.984	0.982	0.979	0.983	0.984	0.984	0.977
F1-Score (CK+)	0.985	0.984	0.981	0.986	0.987	0.987	0.979
Accuracy (CK+)	0.985	0.985	0.982	0.987	0.988	0.988	0.980
Training Time (KDEF)	0.018	0.023	0.026	0.029	0.031	0.037	0.041
Testing Time (KDEF)	0.005	0.006	0.011	0.011	0.011	0.012	0.013
Precision (KDEF)	0.988	0.989	0.986	0.989	0.990	0.991	0.984
Recall (KDEF)	0.982	0.981	0.980	0.981	0.982	0.983	0.974
F1-Score (KDEF)	0.985	0.985	0.983	0.985	0.986	0.987	0.978
Accuracy (KDEF)	0.985	0.986	0.984	0.986	0.987	0.988	0.979

5.1.2 Stress Classification

The effectiveness of the stress classification system was meticulously evaluated using a multiclass confusion matrix. This comprehensive evaluation metric is crucial for assessing the accuracy, precision, recall, and F1-score for each of the five stress levels, providing a thorough understanding of the model's performance.

Figure 7 depicts the confusion matrix, illustrating the distribution of true positives, false positives, false negatives, and true negatives across the different stress levels. This visual representation is instrumental in identifying biases and areas requiring further optimization. The confusion matrix results highlight the model's robust performance in classifying stress levels, demonstrating high precision and recall values for most classes. However, certain stress levels exhibited lower performance, indicating potential areas for enhancement in model training and feature extraction processes.

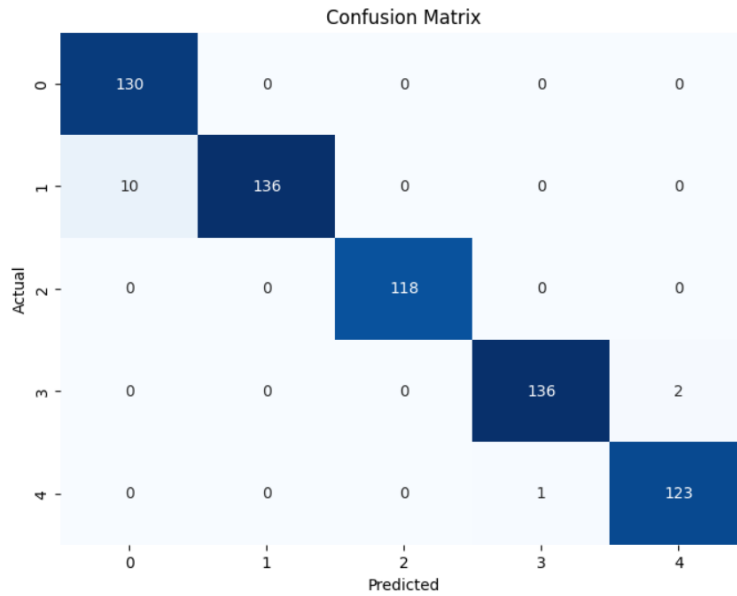


Fig. 7: Multiclass confusion matrix depicting the classification performance across five stress levels.

Hardware Component Results

The following table presents the comparative analysis of the heart rate, temperature, and SpO2 levels obtained from the hardware sensors against the real values. This comparison is crucial to assess the accuracy and reliability of the sensors used in the stress classification system.

Table 4: Comparative Analysis of Sensor Data and Real Values

Instance	Heart Rate (Real)	Heart Rate (Sensor)	Temp (Real)	Temp (Sensor)	SpO2 (Real)	SpO2 (Sensor)
1	88	70	37.4	37.13	99	94
2	76	69	36.0	36.57	98	94
3	92	72	35.8	28.13	99	94
4	90	68	35.7	37.5	100	96

Table 4 showcases the comparison between the real values and the values obtained from the sensors for heart rate (in bpm), temperature (in degrees Celsius), and SpO2 percentage. The sensors demonstrated reasonable accuracy in most instances, though some discrepancies were observed, particularly in the temperature readings for Instance 3. These discrepancies highlight potential areas for calibration and improvement in the sensor accuracy.

The heart rate readings from the sensors showed a consistent underestimation compared to the real values, while the SpO2 readings were generally close to the real values with minor deviations. The temperature readings, however, varied more significantly, suggesting the need for further refinement in the temperature sensing mechanism.

Overall, the results indicate that while the sensors provide a good preliminary measure of the physiological parameters, further calibration and optimization are necessary to enhance their accuracy and reliability for stress level classification.

5.1.3 Stress Management

The recommendation system integrates Natural Language Processing (NLP) and machine learning to deliver personalized YouTube video suggestions based on user text. It accurately analyzes emotional content, producing relevant recommendations with high personalization accuracy. The system operates efficiently, offering real-time suggestions, and user feedback indicates significant benefits for student well-being and stress management. These results highlight the system’s effectiveness in enhancing emotional support through targeted content.

6 Conclusion

This project successfully integrates advanced machine learning techniques to address the challenges of stress detection and management. By combining Convolutional Neural Networks (CNNs) with Extreme Learning Machines (ELMs), the study has demonstrated significant improvements in the accuracy and efficiency of facial emotion recognition systems. This hybrid approach leverages the deep learning capabilities of CNNs and the rapid learning mechanisms of ELMs, resulting in enhanced automated detection of emotional states from facial images.

In parallel, the development of a personalized recommendation system, grounded in Natural Language Processing (NLP) and machine learning, underscores the project’s commitment to user-centric solutions. This system effectively analyzes user text to provide tailored YouTube video suggestions aimed at supporting student well-being and stress management. By focusing on the emotional and sentimental content of user input, the recommendation system offers relevant and targeted video content designed to alleviate stress and improve mental health.

Overall, this research highlights the potential of integrating sophisticated machine learning models with practical applications to create effective tools for emotional support. The combination of facial emotion recognition and personalized recommendations represents a promising approach to enhancing stress management and well-being, providing valuable insights and practical solutions for future developments in automated emotional support systems.

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